Referring through language games

Anonymous ACL submission

Abstract

This paper explores how neural network-based agents learn to map continuous sensory input to discrete linguistic symbols through interactive language games. One agent describes objects in 3D scenes using invented vocabulary; the other interprets references based on attributes like shape, color, and size. Learning is guided by feedback from successful interactions. We extend the CLEVR dataset with more complex scenes to study how increased referential complexity impacts language acquisition and symbol grounding in artificial agents.

1 Introduction

Agents interact with the physical world through their actions and perception, and with other agents through language. Their sensors and actuators allow them to sample the world and their own state using measures that are continuous in nature such as intensity of light, distance, angles, velocity and others which can be measured with a high degree of accuracy. On the other hand, the language that is used to communicate with other agents is based on representations that are composed of a limited set of discrete and arbitrarily chosen symbols. How can both domains and representations arising from these interactions be combined? How are the ranges of measurements expressed in a continuous domain mapped to discrete linguistic labels? How is ambiguity and underspecification resolved? How can agents achieve it through interactive grounding (Regier, 1996; Roy, 2005; Cooper, 2023)?

In this paper we explore how agents based on artificial neural networks learn referential grounding of entities in images of 3-dimensional scenes through language games (Clark, 1996; Bartlett and Kazakov, 2005; Kirby et al., 2008; Steels and Loetzsch, 2009; Zaslavsky et al., 2018). One agent is describing the entities represented as features within bounding boxes of objects, inventing new vocabulary as necessary. The other agent learns to interpret the reference of symbols by identifying one of the bounding boxes based on object attributes such as shape, colour and size. Both agents learn through the success of interaction which is communicated through feedback. The objective we would like us to achive is to model constraints on the model that enable learning language through communication. The novelty of our work, compared with the previous work with this setup (Kharitonov et al., 2019; Lazaridou et al., 2017), consists the extension of the popular CLEVR dataset (Johnson et al., 2017a) with new artificially generated 3-d scenes of objects. These can be referred to based on attributes such as *shape*, *colour* and *size* and discriminated based on different overlaps of these attributes between the target and the distractor objects. Our work is a study of how increasing the complexity of the scene and therefore the space of potential referential expressions to be learned affects learning through interraction of particular configurations of neural networks.

2 Background

3 Dataset

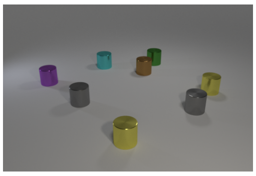
We extend the original CLEVR framework (Johnson et al., 2017a) to have more control over the generated scenes.[1] By this, the objects in the generated images are controlled to have different human-recognizable attributes, namely the *shape*, *size* and *color*. These attributes also correspond to referring expressions in natural language such as English which

[1]Reference to a git repository.

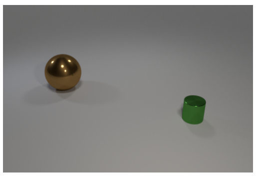
effectively biases the agents to learn a language that is comparable to a human language.

The objects in the scene are separated into two categories: one *target object* and a controlled number of *distractor* objects. The target object is the main object in the scene and the models are trained to identify and communicate between each other. This object is unique in the scene in respect to the attributes. The distractor group contains objects that can share a maximum of two attributes per object. Distractors are not required to be unique.

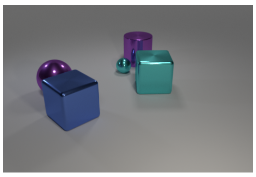
Using these rules, we generate three datasets with the following contraints: (i) the size of the generated images is 480x320 pixels; (ii) 10.000 images are created for each of the datasets; (iii) each image contains a maximum of 10 objects, that are not intersecting, have the same minimum distance between objects and are at least partially visible from the camera.



(a) 'CLEVR color', *small brown cylinder*



(b) 'Dale-2', *small green cylinder*



(c) 'Dale-5', *large purple cylinder*

Figure 1: Example images of each dataset, with the target object specified.

3.1 CLEVR color

The first generated dataset is called 'CLEVR color', in which the target object is identifiable by just the color. Both *shape* and *size* of all distractors are shared with the target object. The distractor group can contain in between 6 and 9 objects.

As seen in Figure 1(a), the *small brown cylinder* is unique. By this, it is possible to refer to the target object using the attributes with four different combinations: the *brown* object, the *brown cylinder*, the *small brown* object and the *small brown cylinder*.

3.2 CLEVR Dale datasets

The above described dataset is very restrictive in the relation between the objects, where only *one* attribute is used to disambiguate them. The number and the type of shared attributes are controlled exactly. In the real world, objects have overlapping attributes and hence objects can often be identified by an intersection of multiple attributes. For this, we created a dataset that allows almost any relation between a target object and the distractors. The creation is inspired by the incremental algorithm for the Generation of Referring Expressions (GRE) described in (Dale and Reiter, 1995) who observe that attributes in descriptions occur in certain order and are added incrementally in a certain hierarchy. This algorithm ensures that every scene contains a unique object in respect to its and the distractors' attributes. Using the algorithm, one can refer to an object using its attributes to discriminate it from all other objects as efficiently as possible. In other words, the object is described unambiguously using the lowest number of words. On the other side, it is not controlled which attributes are shared; they are assigned randomly.

Two datasets following these rules are created. The 'Dale-2' dataset contains one target object and one distractor (see Figure 1(b)), while the Dale-5 dataset contains one target object and exactly four distractors. Consider Figure 1(c), with the target object being the *large purple cylinder*. The large purple sphere shares the size and color, the two cubes only share the size, and the small turquoise sphere doesn't share any attribute.

4 Method

4.1 Image processing

The extract the features and process the images of the datasets, we build upon the proposed architecture in Johnson et al. (2017b) which was used to train baseline models on the original CLEVR dataset. Hereby, the image is first passed through a frozen ResNet-101 model (He et al., 2016). Two convolutional layers with subsequent *ReLU* non-linearities condense the important information from the output of the feature extractor. The convolutional layers reduce the channels to 128 channels, using a kernel size of 3 and a stride and padding of 1.

This matrix represents the encoded image with its extracted features.

4.2 Language Games

The goal of this research is to run and compare different setups of language games systematically. To do this, all experiments rely on the ***E**mergence of lan**G**uage in **G**ames* (EGG) framework (Kharitonov et al., 2019). This framework allows the implementation of language games in code, where two neural models agents communicate through a unidirectional discrete channel. A sender neural model processes visual input. The result is used as the initial hidden state for the encoder LSTM. This LSTM is then producing symbols until it generates an end-of-sequence symbol. The receivers' decoder LSTM processes the message symbol by symbol with a randomly initialized hidden state. The received message sequence is processed symbol by symbol. After each time, a symbol is processed by the LSTM, the resulting new hidden state is passed to the receiver's neural model as the parsed message. The receiver agent is combining it with its representation of the image input and is predicting an output. In other words the receiver agent produces as many outputs as symbols are present in the message. The loss is calculated for each of these outputs separately. These losses are summed up to a total loss that is used to adapt the weights in both agents as well as in both LSTMs. As the discrete sampled categorical distribtion of the message can't be differentiated, we use Gumbel-Softmax relaxation (Jang et al., 2017) to turn it into a continuous distribution, thus allowing backpropagation.

5 Experiments

5.1 Attending in a language game

Setup

In the central experiment, two agents need to solve a referring task together. The receiver needs to 'point' to the target object in the visual scene. However, only the sender is aware of which of the shown objects is the target object. To solve the task correctly the sender is required to generate a referring expression about the target object through the discrete channel while the receiver needs to resolve it. The experiment is set up in a way that avoids explicit human language information as human referring expressions or one-hot encoded attributes. Messages by the sender can only be based on the highly controlled implicit bias in the visual scenes.

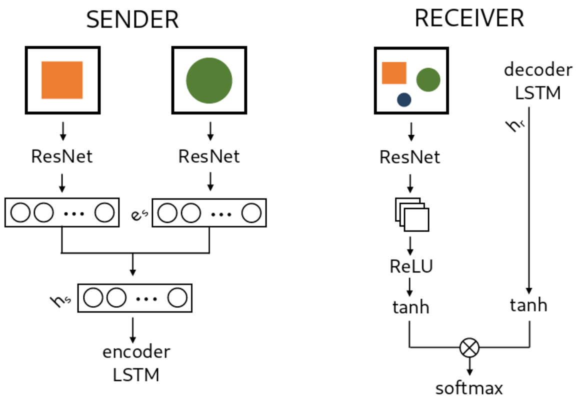


Figure 2: Simplified architecture of the attention predictor game.

Figure 2 shows the simplified architecture of the language game. The sender is given a set of bounding boxes of all objects in the scene, whereas the target object is always the first bounding box and the distractors are shuffled. The features of each bounding box is extracted using ResNet-101 and projected to an image embedding dimension e with a linear layer. All encoded bounding boxes are concatenated and again compressed to the decoder output dimension $LSTM_o$ using another linear layer. This representation of all objects serves as the initial hidden state of an LSTM, which generates the referring expression. Tokens used in the LSTM are embedded with embedding dimension $LSTM_e$. During training, teacher forcing is applied by using embeddings of the ground truth tokens as the input sequence for the LSTM, instead of the output of the LSTM.

The receiver is shown the whole scene including the spatial information. Given the sender's message, the task is to predict the region around target object. For this, the image is divided into 14×14 regions. The target area is located around the center of the target object, consisting of 3×3 regions. The model is then tasked to predict the matrix $A = (a_{ij})$, where:

$$a_{ij} = \begin{cases} 1, & \text{if region } i, j \text{ in target area} \\ 0, & \text{otherwise} \end{cases}$$

The image is encoded using a combination of ResNet-101 and several convolutional layers described in Section 4.1. The resulting matrix has 128×14×14 dimension, corresponding to the 14×14 regions.

The sender's message is decoded using an LSTM with a hidden size $h_r = 500$ and $e_r = 100$. The dot product is calculated between each encoded region of the image and the encoded message, and the $softmax$ function is applied subsequently, which results in a 14×14 matrix. This emphasizes the correlation between the message and each region. A high dot product for a region indicates a high correlation between the message and the specific region, while a low dot product indicates the opposite. Training the agents like this should therefore highlight the regions in the image that are described by the sender, namely the region around the target object. To calculate the loss, the $softmax$ function is applied over the prediction and compared to the ground truth matrix A using *binary cross entropy*.

The experiments are conducted with the following hyperparameters: a learning rate of 2×10^{-4}, a temperature for the Gumbel-Softmax relaxation of 1 and *Adam* (Kingma and Ba, 2015) as optimizer. The following values for the variables are compared:

- $|V|$: 2, 10, 16, 50, 100
- n: 1, 2, 3, 4, 6

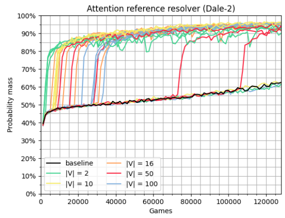
The agents are evaluated on the summed predicted probability for the regions in the target area, the probability mass. In particular, the predicted matrix, consisting of probabilities for each region is multiplied with the ground truth matrix A, consisting of only ones and zeros. The result is summed and returns the probability mass for the target area. If the model predicts the target area perfectly, the probabilities in the target area sum to 1. If the model for instance focuses on the wrong object, the probability mass in the target area is lower.

All results are compared to a baseline in which the sender is generating random messages, so that the receiver needs to solve the task on its own. Any increase in performance requires information being transferred between the agents and the emergence of a language.

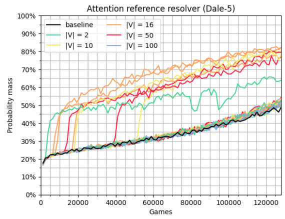
Results

The learning curves are shown in Figure 3. As can be seen, the agents are able to solve the task across all datasets, but with different consistency. However, when the agents start to learn to communicate, the probability mass is boosted instantly to a higher level, where it again learns at a slower speed parallel to the baseline. On the 'Dale-2' dataset, the boost is around 40% points. Most of the learning takes place in the first 40.000 games, but there are also two configurations that increase the performance very late after 70.000 and 105.000 games respectively. Hereby, agents tend to learn faster the smaller their vocabulary size is. The same applies as well to the message length. Using the 'Dale-5' dataset, the probability masses are boosted around 30% points when the agents start to communicate successfully. Compared to the 'Dale-2' dataset, fewer configurations start to converge, while most achieve performances close to the baseline. The smaller number of learning curves makes the analysis more difficult, but the same trend about the vocabulary size and the message length is still visible. Interestingly, only one configuration with $|V| = 2$ beats the baseline, but behaves relatively unstable over the remaining training. On the other hand no configuration with $|V| = 100$ is successful. This indicates that one symbol is too few to encode all meaning, but too many symbols pose a too high difficulty to learn. This hypothesis is amplified by the results on the 'CLEVR color' dataset. Only two configurations beat the baseline, both with a medium-sized vocabulary size and message length. In both cases, the learning takes place relatively late, after 15.000 and 30.000 games respectively.

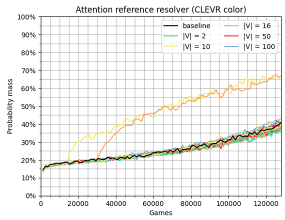
The final probability masses after 128.000 games are summed up in Table 1. Interestingly, the baseline can already find and attend to the correct regions in many cases without the help of the sender. The probability mass is higher than a uniform distribution ($\approx$ 4,6%) and a random guess of an object. It reaches 62,16% on the 'Dale-2' dataset, 49,61% on the 'Dale-5' dataset and 41,68% on the 'CLEVR color' dataset. Looking at the 'Dale-2' dataset, almost all configurations beat the baseline and achieve performances of over 90%, the best



(a) 'Dale-2' dataset with different $|V|$ highlighted



(b) 'Dale-5' dataset with different $|V|$ highlighted



(c) 'CLEVR color' dataset with different $|V|$ highlighted

Figure 3: Learning curves of all attention reference resolver games on each dataset.The colors correspond to different vocabulary sizes $|V|$. The baseline is marked in black.

		Dale-2	**Dale-5**	**color**
n	$\lvert V\rvert$	P **mass**	P **mass**	P **mass**
baseline		62,16%	49,61%	41,68%
2	2	92,27%	52,15%	33,64%
2	10	**96,16%**	80,26%	36,53%
2	16	95,84%	**84,03%**	39,65%
2	50	93,78%	52,24%	39,56%
2	100	92,43%	53,23%	37,68%
3	2	94,52%	51,97%	37,09%
3	10	94,9%	53,47%	38,24%
3	16	94,59%	**81,46%**	**67,88%**
3	50	93,88%	79,65%	40,36%
3	100	95,25%	48,52%	42,55%
4	2	89,15%	51,98%	39,68%
4	10	**96,08%**	48,03%	**64,31%**
4	16	94,14%	49,81%	40,84%
4	50	**96,24%**	48,79%	43,61%
4	100	95,55%	49,65%	42,85%
6	2	59,68%	53,57%	38,43%
6	10	63,46%	**82,12%**	40,11%
6	16	95,86%	50,71%	40,61%
6	50	91,27%	52,55%	40,21%
6	100	60,27%	46,92%	41,98%

Table 1: Probability masses of the attention reference resolver after 128.000 games: n are different maximum message lengths and $|V|$ are different vocabulary sizes.

configurations reach even 96%. Only three configurations stay on the level of the baseline. When comparing the results, mostly the message length n seems to have an influence on the performance. While configurations with $n = 6$ can perform well, this is not constant. All three configurations that don't pass the baseline are allowed to produce message with $n = 6$. $n \in \{3, 4\}$ seem to help the agents the most to perform consistently well, but the difference to configurations with $n \in \{1, 2\}$ is very small. In both cases, the target object is unambiguously identified. The small number of experiments doesn't allow definite conclusions on the influence of the vocabulary size $|V|$, though $|V| = 2$ performs slightly worse than the remaining vocabulary sizes. A correlation between n and $|V|$ is not identifiable.

On the 'Dale-5' dataset, the agents already have bigger problems to beat the baseline. Only 8 out of 30 configuration perform better and reach probability masses around 76% to 84%. However, the increase compared to the baseline is as high as on the 'Dale-2' dataset, with around 30% points. Smaller message lengths ($n \in \{1, 2, 3\}$) as well as a medium-sized vocabulary ($|V| \in \{10, 16, 50\}$) tend to help the agents more, to solve the task successfully. As before, no correlation is visible with the few successful games. That the agents struggle more with the 'Dale-5' dataset is not surprising. First, the larger number of distractors makes it more difficult for the receiver to focus, as can be seen already in the baseline performances. Additionally, in case the sender communicates the attributes of the target object to the receiver, more information needs to be encoded to uniquely describe the target object. This is naturally more complex to learn for the agents. Finally, since more objects are present, they are more likely clustered closer together, which can result in the identification of adjacent regions to the target regions.

The agents struggle the most on the 'CLEVR color' dataset. Hereby, only two configurations perform better than the baseline and reach a probability mass of around 64% to

67%. Both utilize a medium message length of $n \in \{3, 4\}$ and a medium-sized vocabulary of $|V| \in \{10, 16\}$. Interestingly, several configurations with short message lengths of $n \in \{1, 2\}$ perform worse than the baseline. This indicates that there is communication between the agents, but it rather distracts the receiver from the target object towards the distractors. The same argument for a more difficult task when more objects are involved can be made for the 'CLEVR color' dataset. This dataset includes even up to 10 objects present in the scene which increases the likelihood that the receiver focuses on a wrong object.

Appendix A shows examples of the wrongly identified regions on each dataset. These are predictions by the agents that are wrong even thoug a language emerged successfully. Main problems seemed to be target objects not being in the actual frame of the scene that the receiver was processing. However, in several cases (as in the central image), especially for the 'Dale-5' dataset, all objects are visible, and the agents still don't attend solely on the target object. Rather than choosing one of the distractors, the agents usually attend to both objects relatively equally. This indicates that the receiver is uncertain which object the sender is describing. In contrast, the share of errors of the latter type is drastically higher. While a general pattern is difficult to identify, the receiver tends to confuse the target object with distractors that share multiple attributes with each other. In the central and right image, the wrongly identified distractors share both *size* and *shape* with the target object.

5.2 Sender and receiver on their own

Two further experiments are conducted to have a closer look at the sender and receiver models. More precisely, we evaluate how introducing explicit natural language bias into the models changes the training and ability to solve the tasks. This is done by training both the sender and receiver models separately outside a language game context, while the architecture stays the same. Instead of generating and respectively understanding a message of an emergent language, natural language referring expressions are used.

5.2.1 Referring expression generation (sender)

First, instead of generating a message for the receiver, the sender is now tasked to generate natural language referring expressions. The referring expressions for the target object are generated using the incremental GRE-algorithm (Dale and Reiter, 1995). By this, the model needs to describe the target object with respect to the distractor objects.

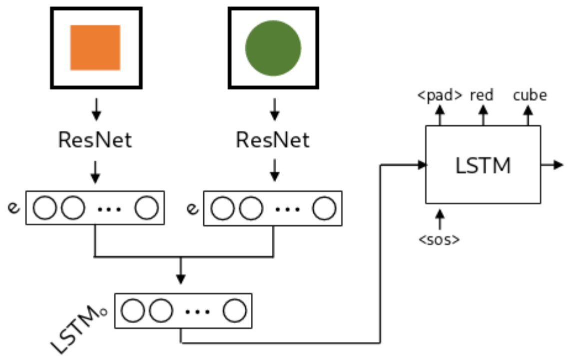


Figure 4: Simplified architecture of the bounding box RE generator

During testing, the LSTM is always forced to generate three tokens, with an embedded start-of-sequence token as first input to the LSTM. Each token in the sequence is determined greedily, by selecting the highest logit in the output of each step in the LSTM. Training is done for 30 epochs and with a learning rate of 2×10^{-4}. The loss is calculated using cross entropy.

This task can be interpreted as a classification task rather than a natural language generation task, as the model is tasked to assign specific attributes to the target object instead of producing free text with a large vocabulary. Following, the model's success is validated on accuracy, recall and precision scores. These are calculated in the following ways. The **overall accuracy** is a measure if the model predicted every word in the referring expression correctly.

Results

Table 2 shows the *overall accuracy*, *F1 scores* for each word. As can be seen, the overall accuracies, in other words perfect matches of the generated referring expression depend very much on the dataset. With the 'Dale-2' dataset, the model can achieve perfect matches in 99%

of the cases in its best configuration. Also the 'CLEVR color' dataset allows the model to predict the correct referring expression in 93% of the samples. Opposed to that the model can only generate perfect referring expressions in 69% of the samples of the 'Dale-5' dataset.

	Accuracy	**F1-Score**
Dale-2	99%	98,57%
Dale-5	69%	89,53%
CLEVR color	93%	95,17%

Table 2: Overall accuracies (Accuracy) and F1-Scores after 30 epochs with embedding size $e = 100$, $LSTM_o = 500$ and $LSTM_e = 30$.

Tables 3 and 4 give a more detailed insight in the results and especially what mistakes the model is making for both the 'Dale-5' and 'CLEVR color' datasets. They list *precision* and *recall* metrics for each token for the best configuration of the model with $e = 100$, $LSTM_o = 500$ and $LSTM_e = 30$. The tokens are grouped by attribute and also show the metrics averaged over each of the attributes. Since the overall results for the 'Dale-2' dataset are already close to perfect, the focus for this analysis lies on the remaining two datasets. The metrics of the *¡pad¿* token indicate if the model produced the correct length of the referring expression, in other words if it was able to determine which attributes are necessary to discriminate the target object from the distractors. For the 'CLEVR color' dataset, the scores are perfect. This is not surprising, because all referring expressions for the 'CLEVR color' dataset consist of exactly two attributes, shape and color, and the first generated token will always be the only *¡pad¿* token in the referring expression (corresponding to the unspecified size). The *¡pad¿* token is therefore easy to learn. For the 'Dale-5' dataset, the model struggles more to predict the correct length of the referring expression.

When looking at the tokens for the shape, it can be seen that the model is able to identify it very well across all datasets. The model predicts the correct shape for all samples using the 'CLEVR color' dataset, while both *precision* and *recall* lie around 98,3% when using the 'Dale-5' dataset. Even though the score is almost perfect, the slight difference might stem from the fact that all distractors have the same shape in the first case, while distractors can be different in the second case. Consequently, the model is only exposed to one shape at a time for each sample, which might simplify its identification.

For the color attribute, the metrics drop significantly for both 'Dale-5' and 'CLEVR color' to an average of around 93%. Hereby, no meaningful difference can be seen across the datasets, but there are differences between the colors. Some colors are predicted with *precision* and *recall* around 95% to 96%, while others are only around 90%. However, these differences are not reproducible across multiple runs and configurations. The best and worst predicted colors vary and no conclusions can be drawn which colors are easier to predict for the model.

Finally, the size is the most difficult attribute to predict for the model. Apart from the 'CLEVR color' dataset, where a size never needs to be predicted and also is never predicted, the metrics for the prediction of size tokens are the lowest across all tokens. They are the only mistakes, the model makes, when exposed to the 'Dale-2' dataset and the average *precision* lies around 23% below the average of predictions of the color for the 'Dale-5' dataset, while the average *recall* lies around 28,82% below. The reason why the *precision* is higher than the *recall* is the *¡pad¿* token, which is predicted very often instead of a token specifying the size. In fact, the opposite relationship is visible for the *precision* and *recall* for said token. The much higher absolute number of *¡pad¿* tokens leads to a smaller relative difference of %-points shown in the table. Again, no conclusion can be drawn if larger or smaller objects are easier to predict, since the results vary across runs and configurations.

In conclusion, it can be said that the model is able to extract discriminative features from the shown bounding boxes and produce referring expressions. However, the result highly depends on the amount of distractors and the resulting need of more discriminative features to describe the target object. While the shape is easily identified, the model has bigger problems to identify the color and especially the size of the target object.

There are two main explanations for these results. First, as already seen in the previ-

		small	large	**size**	cube	cylinder	sphere	**shape**	¡pad¿
Dale-2	Precision	99,17	98,29	**98,73**	99,86	99,71	99,67	**99,75**	99,64
	Recall	97,54	94,26	**95,9**	100	99,56	99,67	**99,74**	99,77
Dale-5	Precision	69,65	69,21	**69,43**	98,19	98,32	98,39	**98,3**	82,22
	Recall	62,11	66,15	**64,13**	98,79	97,87	98,25	**98,3**	84,59
CLEVR color	Precision	-	-	**-**	100	100	100	**100**	100
	Recall	-	-	**-**	100	100	100	**100**	100

Table 3: Precision and Recall in % for ¡pad¿, size and shape tokens with $e = 100$, $LSTM_o = 500$ and $LSTM_e = 30$. The columns **shape** and **size** show the average across all tokens of the respective attribute.

		blue	brown	cyan	gray	green	purple	red	yellow	**color**
Dale-2	Precision	94,51	98,77	97,59	98,68	98,89	98,8	97,47	100	**98,09**
	Recall	97,73	100	98,78	97,4	96,74	98,8	100	98,8	**98,53**
Dale-5	Precision	92,12	93,82	89,13	89,12	92,63	91,12	97,24	94,36	**92,44**
	Recall	92,12	89,78	94,91	94,51	95,71	92,42	89,34	94,85	**92,95**
CLEVR color	Precision	93,46	92,37	94,47	93,86	92,04	91,13	90,07	94,7	**92,76**
	Recall	92,75	92	95,98	89,92	94,12	91,13	94,23	91,91	**92,76**

Table 4: Precision and Recall in % for color tokens with $e = 100$, $LSTM_o = 500$ and $LSTM_e = 30$. The column **color** shows the average across all colors.

ous experiments, a bigger number of distractors confuses the model more where to focus on. In the 'Dale-2' dataset, there are only two possibilities, while the four distractors in the 'Dale-5' dataset and up to nine distractors in the 'CLEVR color' give the model a bigger choice. Moreover, having only one attribute to discriminate the target object, poses the model with more difficulties. The second explanation lies in the incremental GRE-algorithm. For instance using the 'Dale-2' dataset, when only two random objects are placed in a scene, a second (or third) attribute to discriminate the objects is only needed, when the shape is the same. Otherwise, the shape is enough, and the RE is only one word long. The probability for this lies at 66,6%. For shape and color being enough, the probability lies at 29,2% and that all three attributes are necessary is 4,1%. Opposed to that the probabilities with four distractors as in the 'Dale-5' dataset are 19,8%, 47% and 33,2% respectively. With four distractors the algorithm is much more likely to produce longer referring expressions. These are harder to learn and generate for the model since it needs to take more extracted attributes into account to discriminate the target object from the distractors. This is verified by the lower precisions and recalls of the size and color tokens. Furthermore, the low recall of the size tokens shows that the model produces too short referring expressions too often.

5.3 Referring expression resolution (receiver)

As before, the setup of the receiver model stays the same for this experiment, but instead of interpreting the sender's message, natural language referring expressions are passed to the model.

A central problem in the models' predictions is therefore the focussing on precise locations in the image and connecting them to the referring expression. The **attention reference resolver** is meant to solve this problem in two ways. First, the predictions must not be as precise, as only regions in a 14×14 grid need to be identified. Second, the referring expressions are naturally combined with each region in the image. While the *coordinate reference resolver* only combines the complete feature map with the encoded referring expression, the *attention reference resolver* calculates the similarity of each region with the referring expression sep-

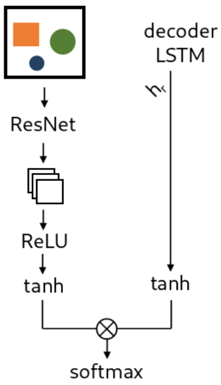


Figure 5: Simplified architecture of the attention predictor

p	Dale-2 P mass	Dale-5 P mass	color P mass
50	94,47%	89,69%	94,36%
100	95,16%	92,19%	95,33%
300	93,65%	93,61%	94,15%
500	96,57%	92,29%	94,15%
1000	95,93%	92,93%	93,82%

Table 5: Probability masses for different projection sizes p of the model after 20 epochs with $LSTM_e = 15$ and $LSTM_o = 1500$

arately. This should connect both encodings easier. As Table 5 shows, this hypothesis seems to be confirmed. Across all datasets, the model is able focus on the correct region in the image with high precision. On the 'Dale-2' dataset, the model is able to make 96,57% of its predictions in the correct regions around the target object with a projection size of $p = 500$. The worst result with $p = 300$ still only lies around 3% points below and therefore identifies the target object almost always correctly. These results still hold, when more distractors and more complex referring expressions are introduced. On the 'Dale-5' dataset, the model can achieve similar performances, with a probability mass of 93,61% with $p = 300$, while three of the remaining configurations stay above 92%. When the *color* is the only discriminating attribute, the model achieves again similar results to the 'Dale-2' dataset, despite the higher number of distractors of up to 10. The best performance of 95,33% is achieved with $p = 100$, but all configurations beat are almost reach 94%. In conclusion, the models are able to easily understand the referring expression and to ground them in the target object in the image. While more complex referring expressions challenge the model slightly more, it still can focus on the correct object.

6 Discussion and Conclusion

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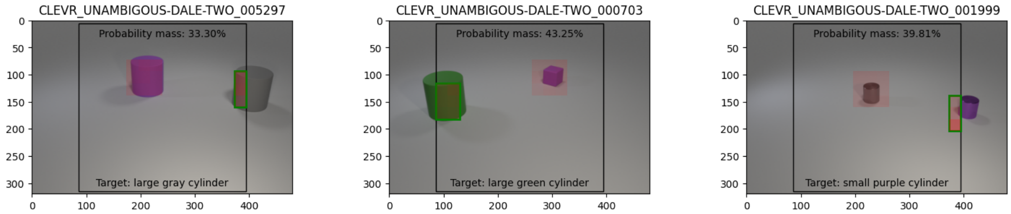
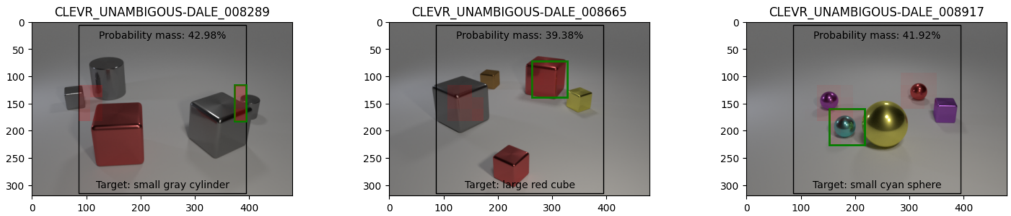
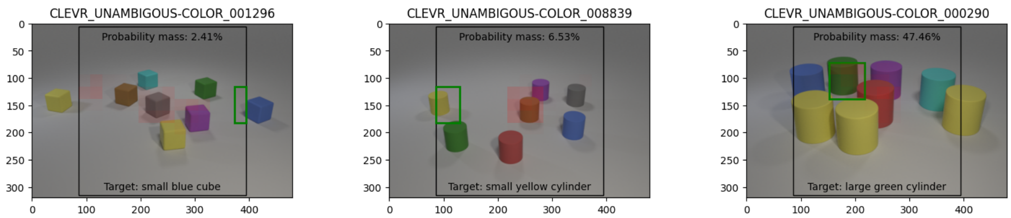
A Examples of errors in attention prediction



(a) 'Dale-2'



(b) 'Dale-5'



(c) 'CLEVR color'

Figure 6: Examples of the *attention reference resolvers'* predictions in games with successful communication with a probability mass lower than 50% on the 'Dale' and 'CLEVR color' datasets. The black rectangle shows the cropped section the model is actually seeing after the image is preprocessed. The green rectangle surrounds the target region that needs to be predicted while the red regions show the actual predictions of the model. The more intense the red, the higher is the probability that the model assigned to this region.